

Numerical Simulation of Train Aerodynamics using a Localized Arbitrary Lagrangian-Eulerian Remeshing Framework in FEniCS

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Abstract

During a train crossing a force can be felt both pushing and pulling the vehicles. Recreating this phenomenon and analyzing the actual pressure changes and force magnitudes is the core objective of this paper. The Navier-Stokes equations are used to simulate the fluid, however since the trains need to move through the mesh, another strategy had to be implemented. This included a localized Arbitrary Lagrangian-Eulerian formulation with periodic topological remeshing allowing the trains to move continuously. The results show three distinct phases during the interaction. As the noses of the trains meet, the pressure between them increases, and they repel each other. When the trains are more parallel, the pressure reaches its lowest point, and the trains pull towards each other. Lastly, as the nose of the train goes past the other tail, the forces and pressure field stabilize to similar values seen from a lone train passing. The peak repulsion force corresponding to lift coefficients of $C_L = +4.25$ (repulsion) and $C_L = -6.0$ (suction). Comparing these results to a real world scenario, gives a resulting 188 kN force over 0.23 seconds, showing how large of a force fluctuation happens within a fraction of a second, explaining the discomfort and wobble felt during this interaction, and also highlights the importance of proper rail and train infrastructure.

1. Introduction

1.1 Background

The operational reality of modern high-speed rail networks frequently dictates that trains pass one another on parallel tracks at extreme relative velocities. When a train moves, it displaces air through its movement. In a crossing scenario, the pressure fields of the trains interact, producing forces orders of magnitude larger than those experienced by a single train. As rail networks continue to push the boundaries of operational speeds, these aerodynamic loads pose significant safety and structural risks. The resulting forces can induce severe passenger discomfort, accelerate structural fatigue on the train body, and generate extreme lateral forces that threaten the operational and directional stability of the vehicles.

This wave of air is already a large force caused by one train by itself. Experiments by Rocchi et al. (2018) show that high-speed trains passing in open air generate a distinct positive pressure wave pushed by the train's nose, which is then followed by a severe low-pressure suction zone along the sides of the vehicle. Clarke and Filippone (2007) further showed that

same-direction passing generates significant pressure interactions in the gap between vehicles.

Fujii and Ogawa (1995) simulated the flow of passing trains in a tunnel and found that there were severe and rapidly fluctuating side forces and yawing moments generated. This comes primarily from the high pressure found at the train noses and the strong low-pressure areas found at the side of the trains. The maximum suction force pulling the trains together occurred as the trains were aligned side-by-side. A similar study by Huang, Li, and Yang (2019) found that the peak pressure coefficients were twice as large when the trains pass each other compared to a single train moving itself. They found the most critical structural region was around the shoulder area of the train. This is because a sole train has air pushed by its nose flow freely outwards. When the trains cross, the air is instead physically obstructed by the other moving train, causing violent acceleration and extreme pressure drops. Specifically they found that, at first when the noses meet, the air is trapped and compressed leading to a slight increase in pressure where the trains repel each other. These results are agreed with what Cui & Zhang (2012) found about passing trains. They also analyzed that a passing interaction has two distinct pulses: the “head wave” and the “tail wave”, where their simulations found that the head waves are significantly larger in magnitude.

Capturing the exact magnitude and phase of these pressure spikes and the resulting forces is critical for safe railway design.

1.2 State of the art

Simulating wind physics on trains has two important aspects to consider. Firstly, the method of capturing wind affecting the train over time, and secondly, the method of moving the trains throughout the simulation.

Previous quasi-steady formulations, using fixed boundaries and time-averaged velocities, are poorly suited for impulsive loads occurring in under 0.5 seconds (Baker, 2010), and diverge rapidly as relative velocity between bodies increases (Jafari et al., 2014), rendering them unsuitable for crossing events. To more accurately simulate moving air, Baker (2010) generated a “virtual” wind time-history. Using known functions of natural wind turbulence and decomposing it into velocity variations with random phases it resulted in an accurate second-by-second representation of the wind speed at the position of the moving train.

To simulate moving the train movement, modern methods use dynamic meshing techniques that come with different significant trade-offs. A sliding mesh technique, utilized by Huang et al. (2019), divides the domain into zones that slide past one another, constantly exchanging data across moving interfaces. Another workaround is the Overset or Chimera Arbitrary Lagrangian-Eulerian (ALE) method. As described by Miller et al. (2014), the Chimera-ALE uses a stationary background mesh and body-fitted grids, allowing the grids to overlap and move independently. This successfully prevents element crushing from global mesh deformation, but has substantial computational overhead from having to transfer and interpolate data between overlapping grids which can lead to numerical dissipation and interpolation errors.

Standard global ALE deforms the entire domain to accommodate structural displacements, which leads to element inversion as the gap between two approaching bodies closes (Miller et al., 2014). To prevent this, Localized ALE approach is used where diffusion equations, such as spring-based smoothing or Laplace solvers, act as numerical “shock absorbers”, distributing the boundary strain smoothly across a localized subset of the mesh (Jafari et al., 2014) (Miller et al., 2014). This methodology provides a tractable implementation within the FEniCS framework while preserving the essential transient physics of the crossing event.

1.3 Research Question

What happens with the pressure field and the forces generated on a train during a train crossing and how do they differ from a train driving alone?

2. Method

2.1 Experiment Setup

The simulations are set up in a rectangular domain Ω in 2D with 3 m in height and 8 m in length. The trains are cutouts of this domain and are set up to be a rectangle of length 2 m and height 0.2 m having a semi-circle at its head with diameter 0.2 m as well. The trains move close to the center of the domain, with 0.4 m separating them. They move in opposite directions at the same speed of 0.1 m/s, and the viscosity of the domain is set to $4 \cdot 10^{-5}$ Pa·s resulting in a Reynolds number of 500, as the density is set to 1 kg/m³. These values do not represent a realistic scenario, but are instead used to create a good simulation environment to show proof of concept. Additionally, the air flow during the simulation is set to zero, only being affected by the moving trains. The specific experiments to be simulated include, a lone train driving, and a train crossing scenario.

2.2 Numerical & Mathematical Model

Because train velocities remain well below the Mach 0.3 compressibility threshold, the flow is modeled using the incompressible Navier-Stokes equations. The boundary conditions are set to be non-slip on the trains, and to allow for free airflow, as opposed to closed walls, open boundaries are set to the edges of the domain. This gives the resulting equations:

$$\begin{cases} \dot{u} + (u \cdot \nabla)u + \nabla p - \Delta u = 0 & x \in \Omega \\ \nabla \cdot u = 0 & x \in \Omega \end{cases}$$
$$\begin{cases} u = u_{boat} & x \in \Gamma_{train} \\ p = 0 & x \in \Gamma_{outer} \end{cases}$$

Where Γ are the boundaries around the train, and the outer walls of the domain. This also leads to still fluid velocity until the train affects it.

Since we need for the mesh inside the boundary to actually stretch to account for train movement, the Arbitrary Lagrangian-Eularian framework is used, as it can capture both the mesh deformation and displacement. We seek a finite element approximation $(u, p) \in V \times Q$ such that:

$$(\dot{u} + ((u - m) \cdot \nabla)u, v) - (p, \nabla \cdot v) + (\nu \nabla u, \nabla v) + (\nabla \cdot u, q) + SD(u, p; v, q) = (f, v)$$

for all test functions $(v, q) \in \hat{V} \times \hat{Q}$ where $SD(u, p; v, q)$ is a residual based stabilization term and the mesh velocity is denoted by m .

These velocity and pressure field solutions are calculated over time using a Crank-Nicolson time stepping scheme, using previous solutions to make a trapezoidal method. At each time step the forces acting on the train in the horizontal (drag) and vertical (lift) direction are calculated. This is done on the boundary of the top train only and is calculated using Green's formula:

$$F(u, p, \phi) = \langle \nu \nabla u \cdot n - pn, \Phi \rangle_{\Gamma} = (\dot{u} + (u \cdot \nabla)u, \Phi) + (\nu \nabla u, \nabla \Phi) - (p, \nabla \cdot \Phi) - (f, \Phi)$$

Where ϕ is set to (1,0) for drag and (0,1) for lift. The final output of forces are in a

$$\text{dimensionless coefficient for drag and lift through } C_L = \frac{2F}{\rho U^2 D} = F \cdot \frac{2}{1 \times 0.1^2 \times 0.2} = F \cdot -1000$$

2.3 Mesh Movement

At each time step, the mesh is deformed in order to track the moving train surfaces. The mesh displacement field u_{mesh} is obtained by solving a Laplace equation.

$$\int_{\Omega} \nabla u_{\text{mesh}} : \nabla v_{\text{mesh}} \, dx = 0$$

Additionally, further boundary conditions are stated surrounding how the mesh will move, and what should stay stationary as seen below. Whereas for the lone train scenario, the bottom train is removed.

$$\begin{cases} \Gamma_{\text{top train}} = v_x * dt \\ \Gamma_{\text{Bottom train}} = -v_x * dt \\ \Omega_{\text{outer}} = 0.0 \end{cases}$$

The outer boundary is defined as an area outside the localized bubble from the trains. This allows for the localized mesh displacement, as opposed to global mesh displacement. This outer boundary is defined as follows:

$$\begin{cases} x_{\min} = \min(x_{\text{top}}, x_{\text{bot}}) - \Delta x, x_{\max} = \max(x_{\text{top}}, x_{\text{bot}}) + \Delta x \\ y_{\min} = \min(y_{\text{top}}, y_{\text{bot}}) - \Delta y, y_{\max} = \max(y_{\text{top}}, y_{\text{bot}}) + \Delta y \\ \Omega_{\text{mesh movement}} = ([x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]) \\ \Omega_{\text{outer}} = \Omega \setminus \Omega_{\text{mesh movement}} \end{cases}$$

With the padding $\Delta x, \Delta y$ set to 0.8 and 0.4 respectively. This results in a large rectangle surrounding both trains being the area where the mesh is allowed to stretch to follow the train boundary movement, while staying connected to the frozen parts of the domain. This method calculates exactly what triangles of the mesh need to stretch and in what direction, in order to prevent element inversion. The mesh velocity is taken by $u_{\text{mesh}} / \Delta t$ which is substituted for m in the ALE formulation.

2.3.1 Remeshing Method

As the mesh continues to stretch from the train movement method, to ensure numerical stability and prevent singularity errors within the discretized Navier-Stokes matrices, a remeshing criterion was implemented based on geometric element quality. At each time step, the solver

calculates minimum radius ratio, being $q = 2 \cdot r_{\text{in}} / r_{\text{out}}$, with r_{in} being the inscribed circle of a triangle of the mesh and r_{out} being the circumscribed circle. A q value of 1.0, is perfect, meanwhile 0.0 means a flat triangle. Therefore, the minimum q allowed is set to 0.15, and below that a remeshing call is made.

When the mesh quality drops below the allowed threshold, the simulation pauses time integration to execute a remeshing sequence. First a new unstretched mesh is created, with the trains at their updated coordinates. Then finite element function spaces, boundary conditions,

and weak forms are recompiled to match the new topology. Finally, the velocity and pressure field solutions are mapped onto the new mesh using a point-wise interpolation scheme.

2.3.2 Post Remesh Force Smoothing

To isolate and mitigate the unphysical pressure spikes induced by local interpolation divergence errors, a dynamic shockwave recovery pipeline was implemented. Upon a remesh event, the solver enters a recovery state for 60 iterations, keeping corrupted force outputs in buffers to prevent data corruption. This window was determined empirically to be sufficient for the pressure field to restabilize following interpolation-induced divergence errors. After this window, a new value is taken from the median value of the final 15 buffered frames. The missing data points within the recovery window are then interpolated using a cubic Hermite blending function $h(p) = 3p^2 - 2p^3$. Where $p \in [0, 1]$ represents the progress through the recovery window. This $h(p)$ is then used as a weight in the function $F_{fill} = (1 - h(p)) * F_{start} + h(p) * F_{end}$, which guarantees a smooth, continuous bridge between the last valid pre-remesh force and the newly settled post-remesh target, eliminating unphysical numerical artifacts while preserving a realistic force history.

3. Results

3.1 Lone Train Simulation

Before analyzing multi-body interactions, a baseline simulation of a solo train moving through ambient still air was conducted to establish reference aerodynamic coefficients. As seen in Figure 1, the velocity field shows the train pushing through the fluid, forcing it forwards and to the sides. As the train moves right, the fluid also fills in the area behind the train. What can be seen from the pressure field is the clear high pressure area at the front, and the two low pressure areas on the shoulders of the train.

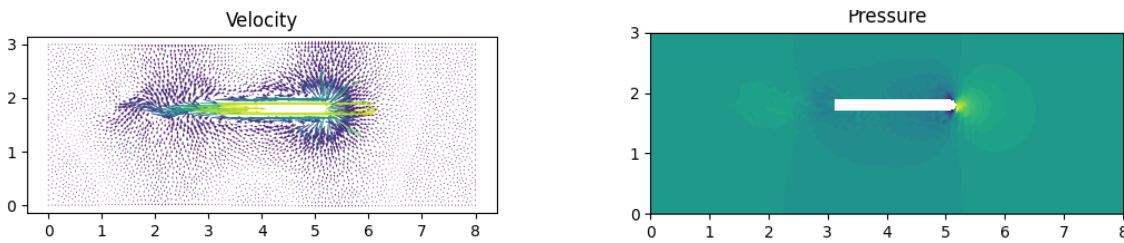


Figure 1: Velocity and Pressure Field of Lone train Simulation at time 21.01

3.2 Train Crossing Simulation

To analyze the coupled aerodynamic interaction, two trains were simulated passing each other in opposite directions at matching velocities. The transient flow evolution is divided into three distinct physical phases.

In the first 'Approach' stage the trains' noses have just passed each other. The high pressure at this part generates higher pressure between them compared to the outside, as seen in Figure 2.

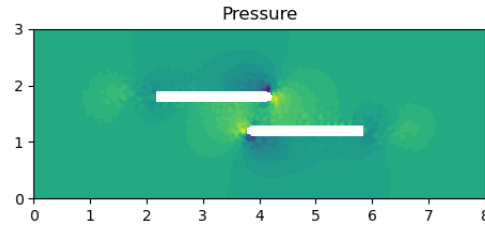


Figure 2: Pressure field in 'Approach' stage at time 11.54

In the next stage, 'Parallel', the trains are completely side by side. Here it is clear to analyze that the pressure between the trains have now switched to being very low compared to outside them, as seen in Figure 3.

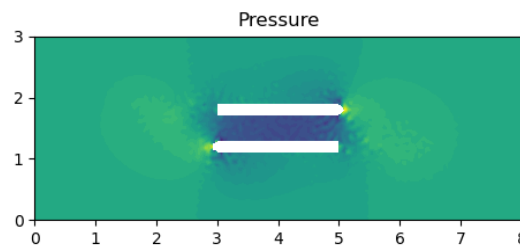


Figure 3: Pressure field in 'Parallel' stage at time 19.96

At the final stage, 'Separation', the trains have just started passing each other. Here the pressure fields between them have normalized and have similarities to what is shown in the lone train simulation, as seen in Figure 4. This separation begins just after the nose has passed the rear of the other train, and continues until the trains have completely left each other and do not affect each other anymore.

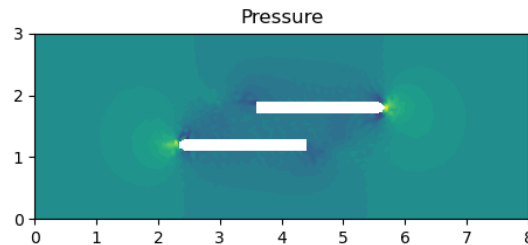


Figure 4: Pressure field in Separation stage at time 25.39

3.3 Comparative Aerodynamic Force Profiles

The analysis of the force coefficients, shown in Figure 5, between the different simulations show a very different picture. For the lone train it is clear that the drag force coefficient stays negative around -0.5 and -0.75 and the lift force coefficient oscillates very close to zero going between around 0.15 to -0.25. This could be because of the train being closer to the top boundary leading to more fluid forces pushing towards the less dense area. The crossing scenario has a similar result with the drag as it stays between to -0.4 and -0.9 during the whole simulation. What is different in the crossing scenario is the lift, as it follows three very clear stages connected to what was stated in 3.2. In the 'Approach' stage at time 11.96 the lift force coefficient peaks at 4.25, and the trains are repulsed by each other. However, as it moves towards the 'Parallel' stage at time 19.46, the coefficient quickly goes negative to -6.0, where

the trains instead attract each other. This peak of -6.0 is approximately 24 times larger in magnitude than the maximum lone-train lift coefficient of 0.25, confirming that the aerodynamic interaction between trains is the dominant loading mechanism. Finally, at the 'Separation' stage the lift forces coefficient on the trains stabilize and start to oscillate around zero.

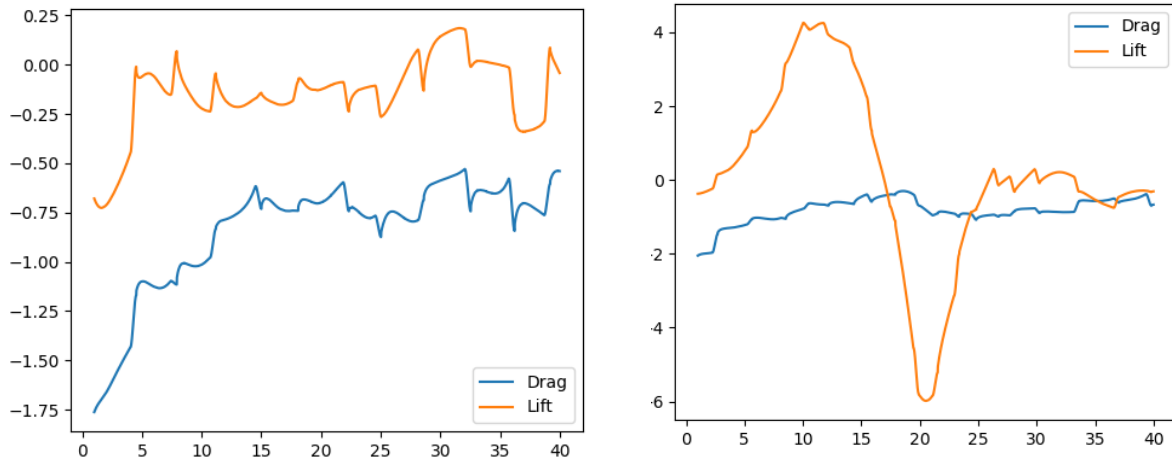


Figure 5: Force comparisons between simulations, lone train (left) and train crossing (right)

4. Discussion

4.1 Fluid Dynamics and Bernoulli's Principle of a Lone Train

The results from the lone train simulation are very logical and easy to understand. The pressure field shows how the train moving forwards creates a high pressure area at the nose, and the fluid around this area is moved to the sides. Rocchi et al. (2018) explains this phenomenon through Bernoulli's principle, as the fluid is forced to accelerate to the sides, whereas velocity increases the pressure decreases, leading to this suction zone at the shoulders.

The forces generated are logical as well. The drag force coefficient staying negative and around the same throughout the simulation is logical as the train moves at a constant velocity leading the fluid continuously pushing in the opposite direction. Similarly, the reason why the lift force coefficient oscillates around zero is because of the Reynolds number causing a Von Karman vortex street to be formed as the train passes the fluid. This results in vortices being generated at the side of the train leading to it being pushed lightly upwards and downwards. These vortices can be seen affecting the trail of fluid behind the train in Figure 1. The reason why it stays at a little bit more negative value compared to positive, is most likely because of the mesh dimensions and train position, where the train is closer to the top boundary which can explain why it gets pushed downwards more.

4.2 The Physics of the Crossing Event

The computed transient force profiles exhibit strong qualitative agreement with established literature. During the crossing scenario I found three distinct phases where the pressure and forces acted very differently. During the 'Approach' stage, the trains' noses have just passed each other, the high pressure area from the noses now interact and the shoulder area where there used to be low pressure, now becomes a place with higher resulting pressure. This leads to the repelling forces seen in Figure 5, which coincides with what Huang, Li, and Yang (2019)

and Cui & Zhang (2011) stated about as the noses meet, the air is compressed leading to higher pressure and pushing the trains away from each other.

Consistent with Fujii and Ogawa (1995), the maximum suction force occurs as the trains are fully parallel. My 'parallel' phase clearly represents this as well, as a large low pressure area is found between the trains, and the resulting forces lead to the trains being pulled together. Cui & Zhang (2011) explains this by saying that when the high pressure noses have passed the low pressure shoulder regions, the area between the trains results in a low pressure region. When the trains overlap, the narrow gap between them acts like a Venturi nozzle, where the fluid is unable to escape laterally. The relative velocity of the train walls bounding this narrow channel is $V_1 + V_2$. The air trapped in the gap is sheared by both moving walls simultaneously, forcing the fluid velocity within the gap to skyrocket far beyond the speed of either individual train. Because the fluid velocity in this restricted channel is massively amplified by the relative motion of the train walls, the static pressure inside the gap plunges to its absolute minimum value. Meanwhile, the pressure on the outside of the trains remains close to atmospheric pressure. This severe pressure asymmetry pushes the train bodies inward toward each other with maximum force (Huang et al., 2019).

During the 'Separation' stage, Cui & Zhang (2011) found that a separate 'tail-wave' was found to distinctively affect the trains. The simulation did not reproduce the tail wave. This is likely attributable to two factors. First, the lift force integrates over the entire train surface simultaneously, cancelling local rotational effects. Secondly, the 2D constraint and lower Reynolds number than reality prevents vertical pressure dissipation, which in reality moderates the tail separation dynamics. Instead the pressure fields stylised quickly and showed similarities to the lone moving train, as seen in figure 1.

Similarly, another surprising result was the drag force coefficient during the crossing scenario. This value did not sway as much as expected as the velocity fields from the trains interacted with each other. Instead it stayed consistent, closely resembling the lone train scenario with a little bit more volatility. The drag force reflects the difference in pressure between the front and back of the train, as the noses of the trains pressure fields never changed in a distinct way, and the same for the tails, it leads to the drag force coefficient not changing largely. Instead the pressure changes were largest on the lateral sides of the trains, affecting the lift force a lot more.

4.3 Analysis of the forces

The computed peak suction during the simulation corresponds to a lift coefficient C_L of 6.0, while the peak repulsion gives a lift coefficient C_L of 4.25. To evaluate the structural impact on a full-scale scenario, these coefficients are scaled using the standard aerodynamic force equation:

$$F_{real} = C_L * \frac{1}{2} \rho U^2 A_{ref}$$

Where the density of air is 1.225 kg/m^3 , the speed is 50 m/s , and the reference area is the frontal area of the train estimated to be 12 m^2 (from 3 m width and 4 m height). Applying these values results in a real suction lift force of approximately 110.25 kN , where the peak repulsion is 78.1 kN . The critical observation is the rate of change of these lateral loads. The peak repulsion happens just after the trains' noses have passed each other at the combined nose length of 0.2 m , ending with peak suction as they are fully parallel, meaning at the distance of 2.1 m . Scaling this to a 25 meter train gives a distance of 22.6 m traveled, where the train's combined speed is $50 \text{ m/s} + 50 \text{ m/s} = 100 \text{ m/s}$, gives a total time passed of $\Delta t = 0.226 \text{ seconds}$.

During this brief interval, the aerodynamic lateral load transitions from +78 kN of repulsion to -110 kN of suction, a net swing of approximately 188 kN. This is equivalent to roughly 19 tons of lateral force rapidly reversing direction, highlighting the potential for significant fatigue accumulation in rail infrastructure.

This calculated value can not be said to be truly accurate and instead gives an upper bound of possibilities. This over-prediction is a fundamental limitation of utilizing a 2D domain to model a 3D phenomenon. In a two-dimensional plane, the fluid is artificially constrained and all displaced air is forced to compress strictly between the two passing bodies, leading to massive localized pressure gradients compared to reality where air can escape vertically. Also using the frontal area for lateral force is not completely logical. This was done since the code normalization for the lift coefficient already uses the frontal length, therefore this is done in this scaling to stay consistent.

5. Conclusion

In conclusion, the pressure field during a train crossing goes through three distinct phases. During the 'Approach' stage, the trains are repelling each other as the high pressure of the noses have just passed each other. When the trains are later parallel, the largest force generation is seen as a large low pressure area is created between the trains fully pulling them together. Finally, after the noses of the trains have passed the other tail, the forces and pressure around them start to stabilize. This final 'Separation' stage is what is most closely resembled to a lone train driving, and is where it is most clear that there is a high pressure area at the nose, with a lower pressure area found at the shoulder area. This can be explained through Bernoulli's principle as the fluid is forced to accelerate. Comparing the force coefficients to the lone train scenario shows a much calmer scenario with lift force coefficients being more than 20 times smaller compared to the peak repulsion and suction seen from a crossing. Surprisingly, the drag force coefficient is generally unaffected by the crossing scenario and stays at a consistent value throughout the simulation, similar to the lone train scenario. Scaling the results to real world values gives a force change of 188 kN over a duration of 0.226 seconds. This shows just how large the forces are interacting during this crossing, and why this is so important to consider when building railroads to ensure passenger safety.

6. Future Work

While the developed computational framework successfully demonstrates the viability of modeling multi-body transient aerodynamics using a localized ALE approach, several natural extensions are proposed to enhance physical realism. First, the solver should be extended from a two-dimensional plane to a fully three-dimensional domain. In a 2D constraint, fluid displacement is artificially compressed laterally. A 3D model would allow air to escape over the train roofs, yielding highly accurate, comparable aerodynamic coefficients, which was also stated as a limitation and something affecting results heavily by Clarke & Filippone (2007).

Second, future work will move beyond simple surface force integration and pressure field to analyzing the triple decomposition of shear, strain and rotation of the pressure field, and force generation at more specific points of the trains. This would allow for a better understanding of what is happening with the pressure fields and why, and seeing further rotational forces acting

on the train, as lift forces can be high in one area but low or even negative at another during the same time frame. This could help find the 'tail wave' analyzed by Cui & Zhang (2011).

Finally, changing the method to fully implementing true overset (Chimera) Grids, as done by Miller et al. (2014). While my solution allows for the train to move without inverting elements, it requires periodic remeshing. This results in misses in force generation, and spikes in the pressure fields affecting the simulation. Upgrading to support true overlapping overset grids would allow the train meshes to slide past each other infinitely without a single element ever deforming, completely eliminating the interpolation pressure spikes.

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